

Augmenting Web Clipping Applications with C/C++

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Objective

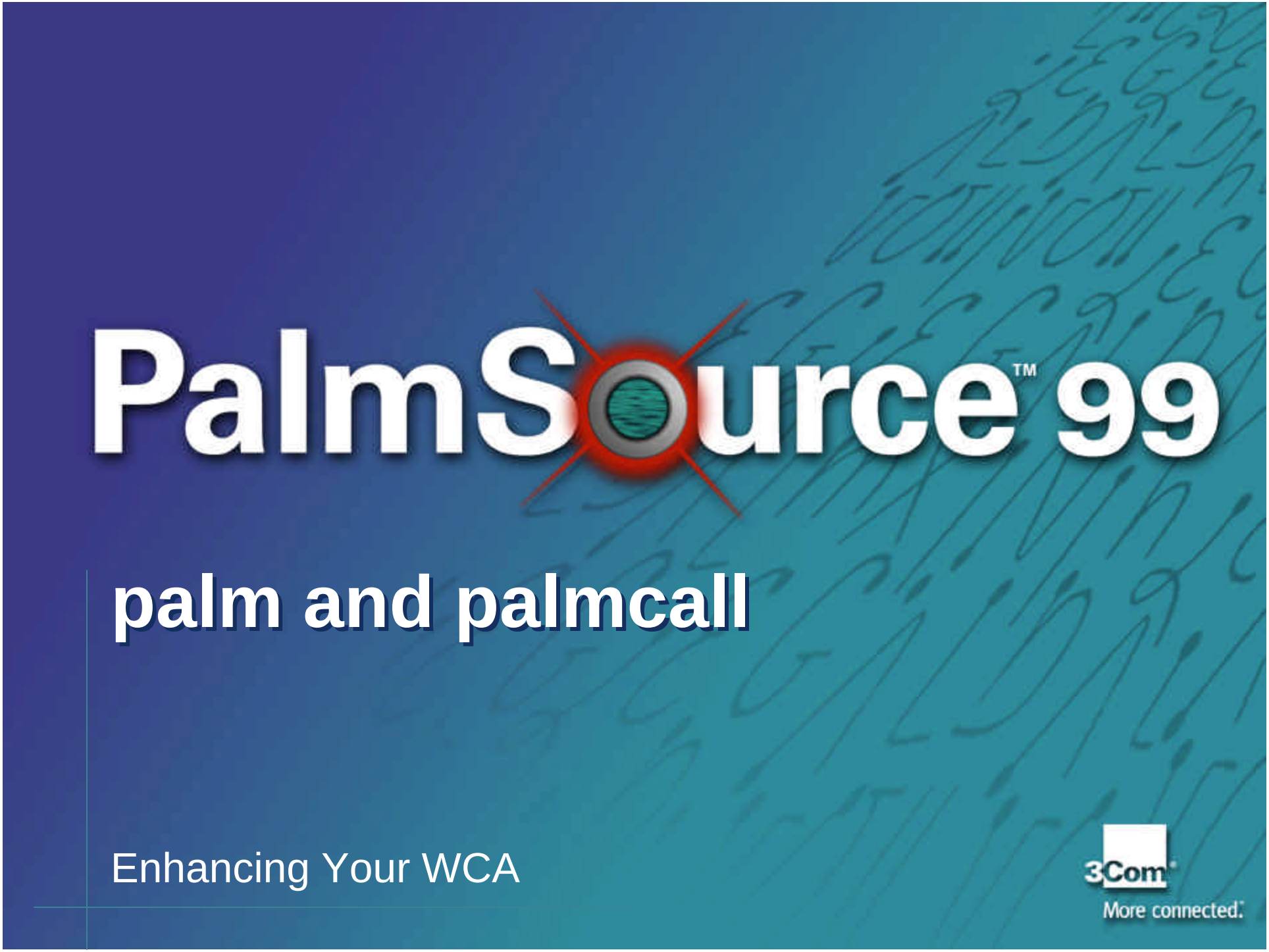
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- To show how web clipping applications can be integrated with Palm™ C/C++ applications
  - Enhancing a WCA with a C/C++ application
  - Enhancing a C/C++ application with a WCA

Agenda

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- palm and palmcall
  - Enhancing your WCA
  - Parsing input
  - Displaying your UI
  - Modifying your database
- Enhancing your .PRC
- Conclusion
- Q&A


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palm and palmcall
Enhancing Your WCA
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palm and palmcall

- What are palm and palmcall?
  - Method by which WCAs can launch and send data to other applications
- What can I do with palm and palmcall?
  - Pseudo-cgi on your handheld
  - Process data from web clipping pages
    - e.g., graphing, updating databases, etc.
  - Store data on the handheld for web clipping applications
    - e.g., preferences

palm

- **`palm:XXXX.appl?YYYY`**
  - XXXX = Application Creator ID
  - YYYY = parameters
- Clipper exits and your application opens
  - **`sysAppLaunchCmdNormalLaunch sent`**
  - Pointer to URL-string sent as parameter block
    - e.g., **`“palm:funA.appl?par1=val1&par2=val2”`**

palm

Supporting palm in your application

- Check for launch codes
  - **`sysAppLaunchCmdNormalLaunch`**
- Parse parameters in URL-string (if any)
- Do application's stuff

palm

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Calling from a Web Clipping application

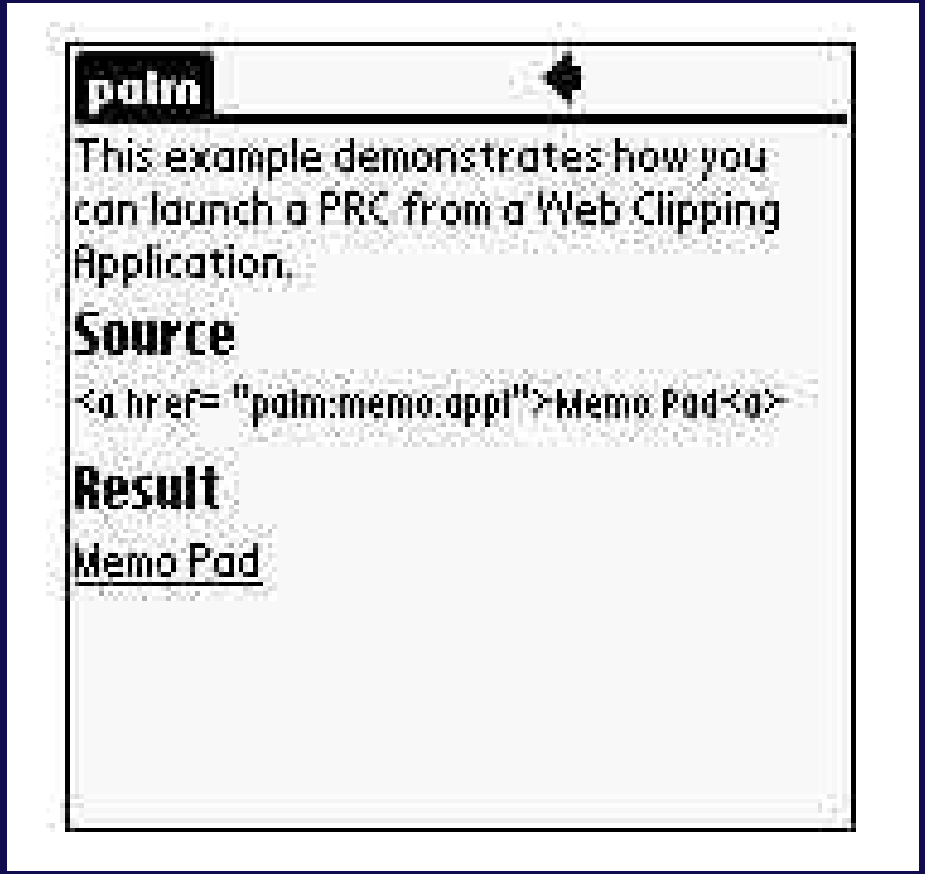
- Anchor Tag
  - `<a href="palm:funA.appl?type=Hello!">Have Fun</A>`
- Form

```
<form method="get" action="palm:funA.appl">
        <input type="text"   name=”type">
        <input type="hidden" name="ver" value="1"
        <input type="submit" value=”Have Fun">
</form>
```

palm

Demo: palm example

- palm



palmcall

- **`palmcall:XXXX.appl?YYYY`**
  - XXXX = Creator ID
  - YYYY = parameters
- Clipper sublaunches your application
  - **`sysAppLaunchCmdURLParams sent`**
  - Pointer to URL-string sent as parameter block

palmcall

Restrictions and considerations

- No access to global variables
- Return control back to Clipper

palmcall

Supporting palmcall in your application

- Check for launch codes
  - **`sysAppLaunchCmdURLParams`**
- Parse parameters in URL-string (if any)
  - e.g., in **`"palmcall:aMEM.appl?memo=Hello!"`**

    `“memo=Hello!”`
- Do application’s stuff
- Return to Clipper
  - e.g., add a stop event to the event queue

palmcall

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Calling from a Web Clipping application

- Anchor Tag
  - `<a href="palmcall:aMEM.appl?memo=Hello!">Add</A>`

- Form
  - ```
    <form method="get" action="palmcall:aMem.appl">
          <input type="text"   name="memo">
          <input type="hidden" name="ver" value="1"
          <input type="submit" value="Add">
    </form>
    ```
    ```

palmcall

Simplified application example code:

```
DWord PilotMain( Word cmd,
                 Ptr  cmdPBP,
                 Word flags)
{
    switch(cmd) {
        case sysAppLaunchCmdNormalLaunch:
            // normal launch or palm launch
            break;
        case sysAppLaunchCmdURLParams:
            AddMemo(cmdPBP);
            break;
    }
    return 0;
}
```

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Parsing Input

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Parsing Input

- Algorithms
  - Many methods and theories for parsing
  - Needn't be complex
  - Should handle exceptions gracefully (as always)
- Format of URL-string
  - "palmcall:XXXX.appl?A=v1&B=v2&C=v3..."
  - You decide parameter format (anchor tag); e.g.,
    - A=v1&B=v2…
    - A&B&C…
    - va1,va2,va3&vb1,vb2&vc&…
  - Form tag will implement URL parameter format
    - "palmcall:xxxx.appl?A=v1&B=v2&C=v3..."

Parsing Input

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- Hexadecimal representation
  - Some characters will be converted to their hexadecimal values; e.g.,
    - “http://www.tow.com/” to
    - “http%3A%2F%2Fwww.tow.com”
- sysAppLaunchCmdGoToURL
  - You needn’t provide URL strings with hex characters
  - Beware of back-end server requirements

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Displaying Your UI

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Displaying Your UI

- You can display forms from within a web clipping application with palmcall
- Remember, no access to global variables
- Must return control back to Clipper
  - e.g., stop the AppEventLoop by adding a stop event to the event queue

```
EventType evt;
evt.eType = appStopEvent;
EvtAddEventToQueue(&evt);
```

Displaying Your UI

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Simplified example code:

```
// Save the web clipping application form
origForm =  FrmGetActiveFormID ();

// Initialize the dialog
frm = FrmInitForm (MainForm);
FrmSetEventHandler (frm, MainFormHandleEvent);

// Display the dialog
FrmSetActiveForm (frm);
FrmDrawForm (frm);

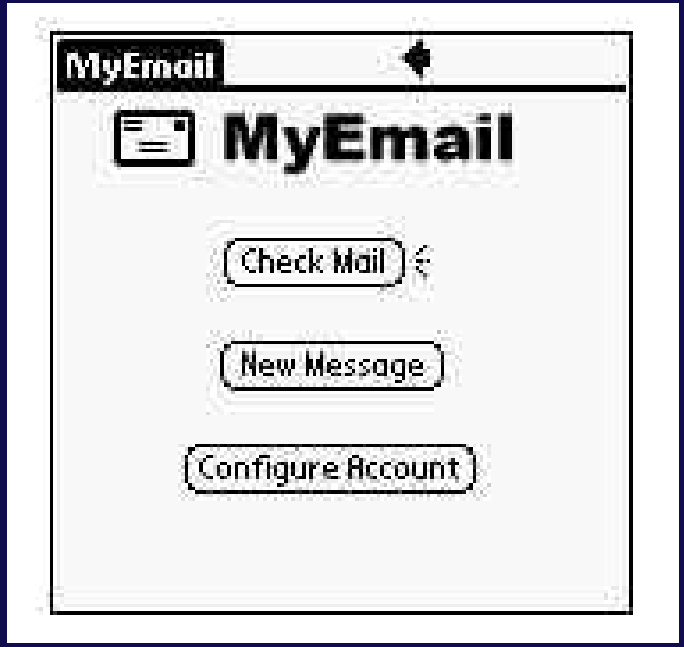
// Run your application
AppEventLoop();

// Erase the dialog and return to Clipper
if (origForm)
  FrmReturnToForm (origForm);
```

Displaying Your UI

Demo: MyEmail

- palmcall
- New characters
- **`sysAppLaunchCmdURLParams`**
- Parsing input
- Displaying UI



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Modifying Your Databases

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Modifying Your Databases

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- Format of URL-string
  - "palmcall:Mema.appl?memo=X"
- AddMemo Routine
  - Parses input for "memo="
  - Creates new memo record

Modifying Your Databases

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```
DWord PilotMain(Word cmd, Ptr cmdPBP, Word launchFlags)
{
        Err error;
        switch (cmd) {
                case sysAppLaunchCmdNormalLaunch:
                        break;
                case sysAppLaunchCmdURLParams:
                        if(cmdPBP) {
                                error = AddMemo(cmdPBP);
                                if(error) FrmAlert(ErrorAlert);
                                else FrmAlert(MemoSuccessAlert);
                        }
                        break;
                default: break;

        }
        return 0;
}
```

Modifying Your Databases

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```
static Err AddMemo(CharPtr cmdPBP)
{
    DmOpenRef     MemoDB;
    LocalID       MemoDBID = DmFindDatabase(0, "MemoDB");
    Word          index = 0;
    VoidHand      newMemoHandle;
    Ptr           newMemoPtr;
    CharPtr       memo;
    int           memoLength;

    memo = GetParameter(cmdPBP, kMemoParamStr);
    if(!memo) return -1;

    memoLength = StrLen(memo) + 1;
    if(memoLength <= 1 || !MemoDBID) return -1;

    // Open the Memo Records Database
    MemoDB = DmOpenDatabase(0, MemoDBID,dmModeReadWrite);
```

Modifying Your Databases

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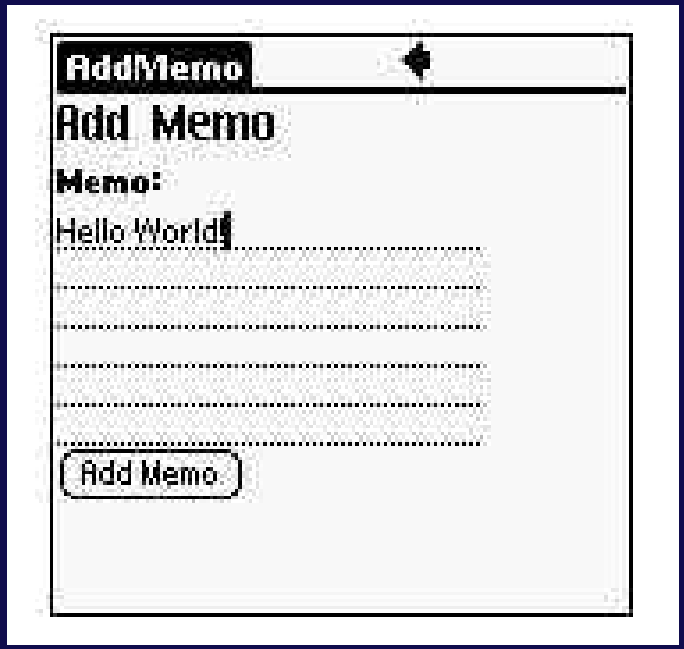
```
    if (MemoDB) {
            newMemoHandle = DmNewRecord(MemoDB, &index,
                                        memoLength);
            if(newMemoHandle) {
                    // Create the new memo record
                    newMemoPtr = MemHandleLock(newMemoHandle);
                    DmWrite(newMemoPtr, 0, memo, memoLength);
                    MemHandleUnlock(newMemoHandle);
                    DmReleaseRecord(MemoDB, index, true);
                    MemoSortRecord(MemoDB, &index);
                    DmCloseDatabase(MemoDB);
                    // Success!
                    return 0;
            }
    }
    return -1; // Failure
}
```

Modifying Your Databases

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Demo: Add Memo

- palmcall
- **`sysAppLaunchCmdURLParams`**
- Parsing input
- Adding data to a database



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Enhancing a C/C++ App

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Enhancing a C/C++ App

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- The Palm VII™ SDK
- Constants
- Key Codes
- Characters
- Launch Codes
- Wireless Internet Feature Set

Enhancing a C/C++ App

Constants

- **`sysFileCClipper`**
  - Creator ID for Clipper application
  - **`'clpr'`**
- **`sysFileTPQA`**
  - File type for web clipping applications
  - **`'pqa '`**
- **`sysFileCMessaging`**
  - Creator ID for iMessenger™
  - **`'msgs'`**

Enhancing a C/C++ App

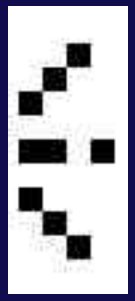
Key Codes

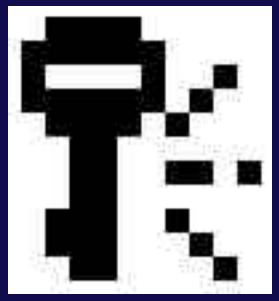
- **`vchrHardAntenna`**
  - Antenna has been raised
  - Trap this key event in your AppEventLoop to prevent the unit from going to the Launcher
- **`vchrRadioCoverageOK`**
  - Unit is within coverage of a base station
- **`vchrRadioCoverageFail`**
  - Unit could not find a base station

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Characters

- Use these characters to inform the user whenever a wireless transaction will occur
  - **`chrOta`**
  - **`chrOtaSecure`**





Enhancing a C/C++ App

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Launch Codes

- **`sysAppLaunchCmdGoToURL`**
- **`sysAppLaunchCmdOpenDB`**
- **`sysAppLaunchCmdAddRecord`**
  - For iMessenger™ (Mail as well)

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sysAppLaunchCmdGoToURL

- From app to Clipper
- Instructs Clipper to display the requested URL
- Parameter block is a pointer to a URL-string
  - e.g.,
    - **“http://www.palm.com/devzone/palmvii”**
    - **“file:your.pqa/index.html”**
- Currently accepted protocols
  - http://
  - https://
  - file:
  - palm:
  - palmcall:

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Simplified example code:

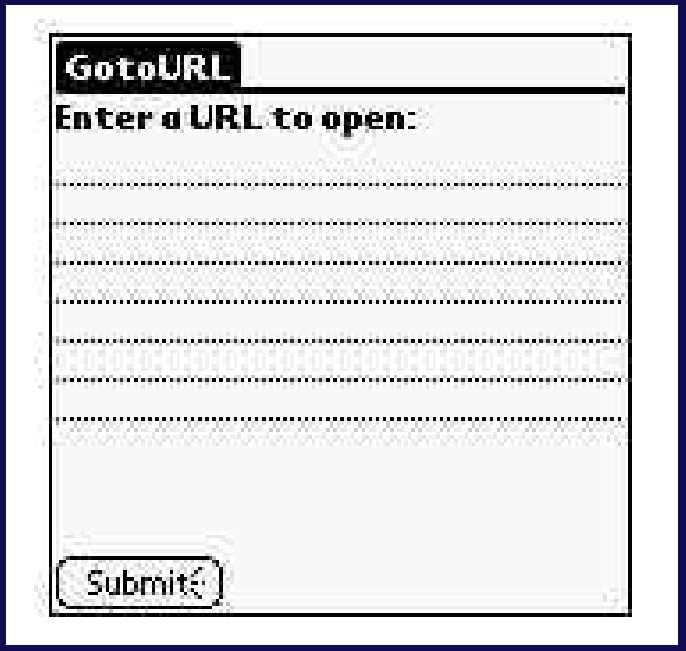
```
// set owner of urlP to OS:
MemPtrSetOwner(urlP, 0);

// launch Clipper and go to the url:
SysUIAppSwitch(cardNo,
               ClipperDbID,
               sysAppLaunchCmdGoToURL,
               urlP);
```

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Demo: GotoURL

- New characters
- Parsing input
- **`sysAppLaunchCmdGoToURL`**



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sysAppLaunchCmdOpenDB

- Instructs Clipper to open the specified WCA
  - Best to use **sysAppLaunchCmdGoToURL**
- Parameter block is a pointer to a **SysAppLaunchCmdOpenDBType**

```
typedef struct {
            Word     cardNo;
            LocalID  dbID;
} SysAppLaunchCmdOpenDBType;
```

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Simplified example code:

```
sysAppLaunchCmdOpenDBType* wcaInfo =
   MemPtrNew(sizeof(sysAppLaunchCmdOpenDBType));

// assign info for WCA to open:
wcaInfo->cardNo = 0;
wcaInfo->dbID   = DmFindDatabase(0,
                                 “your.pqa”);
// set owner of wcaInfo to OS:
MemPtrSetOwner(wcaInfo, 0);

// launch Clipper and your.pqa:
SysUIAppSwitch(cardNo,
               ClipperDbId,
               sysAppLaunchCmdOpenDB,
               wcaInfo);
```

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sysAppLaunchCmdAddRecord

- Launch iMessenger™ from your WCA or .PRC
- Parameter block is a pointer to a **`MsgAddRecordParamsType`** structure:

```
typedef struct {
            Word    category;
            Boolean edit;
            Boolean signature;
            CharPtr subject;
            CharPtr from;
            CharPtr to;
            CharPtr replyTo;
            CharPtr body;
} MsgAddRecordParamsType;
```

Enhancing a C/C++ App

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Simplified example code:

```
MsgAddRecordParamsType *msg;
msg = MemPtrNew(sizeof(MsgAddRecordParamsType);

msg->edit       = false; // true displays editor
msg->signature  = false;
msg->replyTo    = NULL;
msg->from       = NULL;
msg->category   = 1;
msg->body       = "Welcome to Palm Source 99!"
msg->to         = "Scot_Stennis@palm.com";
msg->subject    = "Welcome";

MemPtrSetOwner(msg, 0);
SysAppLaunch(cardNo, iMessengerDbId, 0,
             sysAppLaunchCmdAddRecord,
             msg, &result);
```

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sysAppLaunchCmdAddRecord



Enhancing a C/C++ App

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Wireless Internet Feature Set

- Check for Clipper or iMessenger™ before using any wireless Internet features:
  - ```
    Err                 err;
    DmSearchStateType   searchState;
    UInt                cardNo;
    LocalID             dbID;

    err = DmGetNextDatabaseByTypeCreator(
                true,
                &searchState,
                sysFileTApplication,
                sysFileCClipper, // or sysFileCMessaging
                true,
                &cardNo,
                &dbID);
    ```
  - An error is returned if not present
    ```

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Conclusion

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Summary

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- Use C/C++ to augment your web clipping applications
  - palm and palmcall
  - **`sysAppLaunchCmdURLParams`**
- Augment your C/C++ apps with wireless features
  - Constants
  - Key codes
  - Characters
  - Launch codes
    - **`sysAppLaunchCmdGoToURL`**
    - **`sysAppLaunchCmdOpenDB`**
    - **`sysAppLaunchCmdAddRecord`**

What Next?

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- Session 481 Lab
  - Specifically for Palm VII™ development
  - See sample code and do a quick exercise
- Session 450
  - Learn wireless INetLib APIs for the Palm VII
    - Raw throughput of data
    - Wireless Internet access without web clipping applications
- Palm VII SDK available in Code Warrior for Palm Computing® Platform, Release 6

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Questions and Answers

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